

- 6 (a) Fig. 6.1 shows a transmission electron micrograph of a section through a mitochondrion.

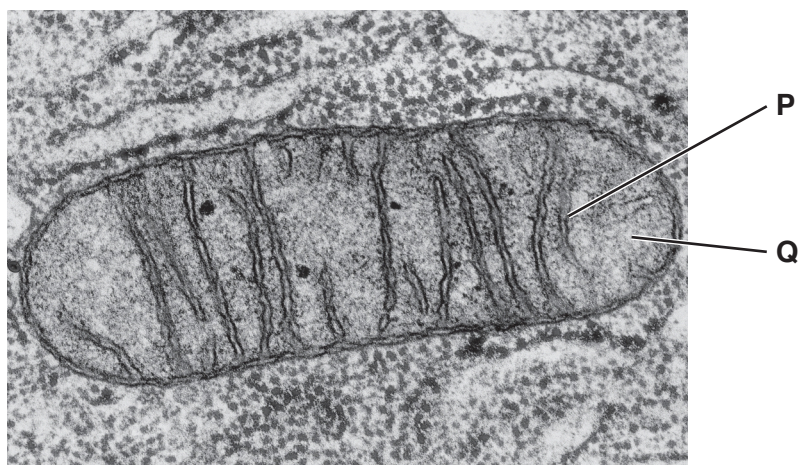


Fig. 6.1

Table 6.1 shows some structures and compounds involved in aerobic respiration.

Use letter **P** or letter **Q** from Fig. 6.1 to complete Table 6.1 to show the location of the structures and where the compounds are used.

Table 6.1

compound or structure	location
ATP synthase	
acetyl CoA	
phospholipid bilayer	
oxaloacetate	

- (b) The number of mitochondria differs from cell to cell. Cells also vary in size and volume.

Table 6.2 shows the mean number of mitochondria per cell, the mean cell volume and the mean number of mitochondria per μm^3 of three types of mammalian cells.

Table 6.2

cell type	mean number of mitochondria per cell	mean cell volume / μm^3	mean number of mitochondria per μm^3
fat cell	100	600 000	0.17×10^{-3}
heart cell	2000	45 000	44.44×10^{-3}
liver cell	2000	125 000	

- (i) Calculate the mean number of mitochondria per μm^3 for the liver cell.

Write your answer in Table 6.2.

[1]

- (ii) Suggest reasons for the difference in mean number of mitochondria per μm^3 between the fat cell and the heart cell.

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[2]

- (iii) Suggest why it is difficult to compare the ability of the three types of cell to respire aerobically, based **only** on the mean number of mitochondria per cell.

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[1]

[Total: 7]